

11 Eigenvalue Decomposition

11.4 The Gram Matrix

from inner products and L2 norms to spectral decomposition, orthogonalization, and the SVD



Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills You Will Master in This Section

1

A Gram matrix is symmetric, and positive semidefinite (or positive definite)

2

How the Gram matrix relates to inner products of the original matrix's columns

3

Main-diagonal entries equal the squared L2 norm of each column vector

4

How to spectrally decompose a Gram matrix

5

Using an orthogonal matrix to orthogonalize the original matrix

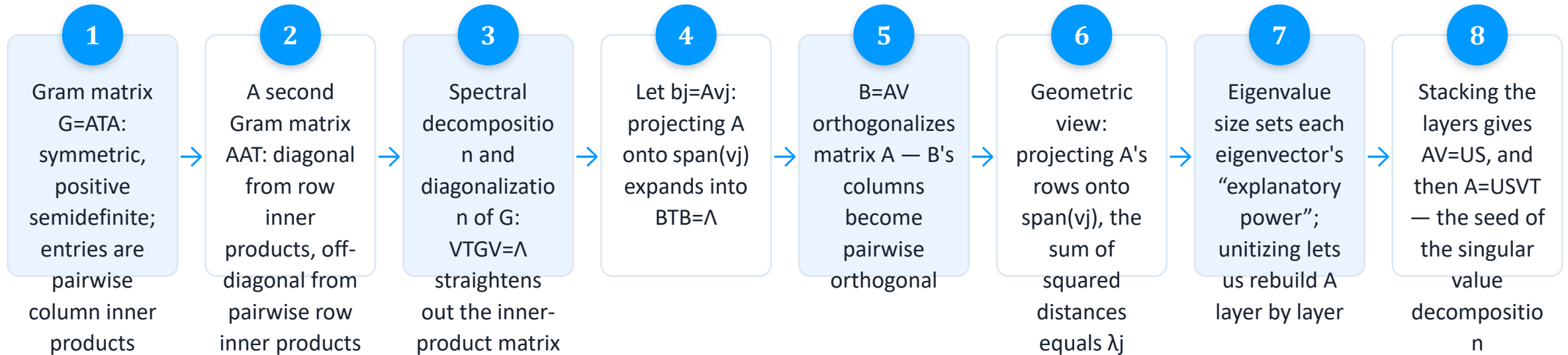
6

Ranking eigenvectors by “explanatory power” from their eigenvalues

7

How the Gram matrix's spectral decomposition connects to the SVD

From $G=ATA$ to orthogonalization, then to the singular value decomposition



DEFINITION

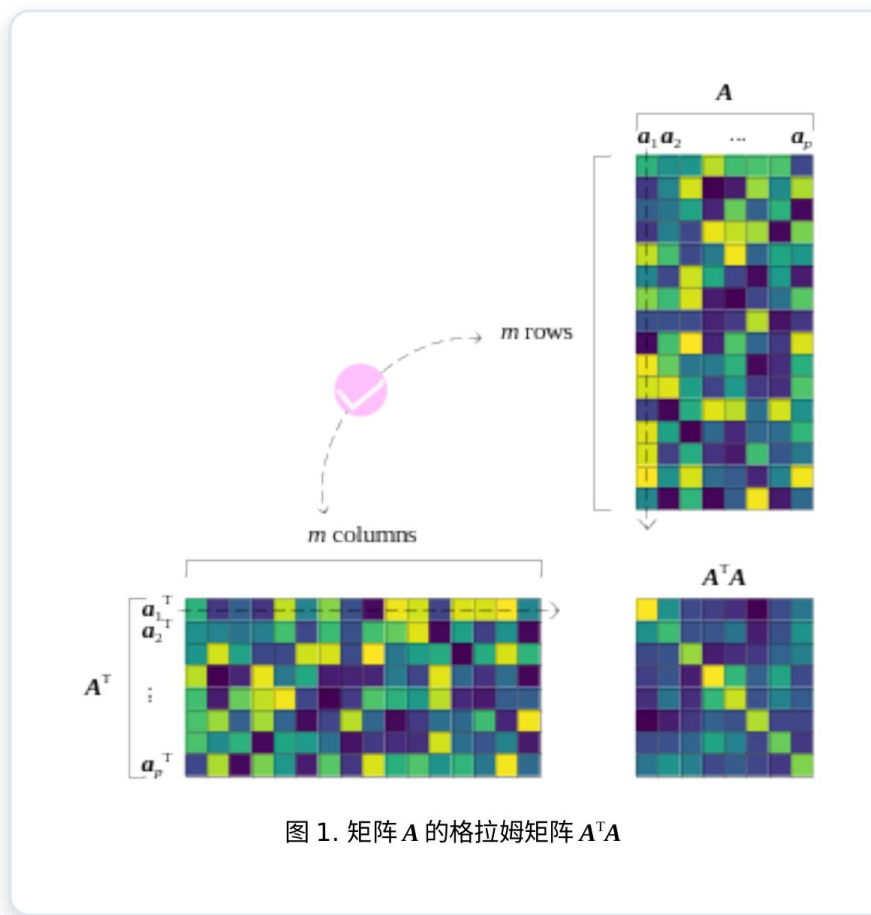
Given matrix A with shape $m \times p$, the Gram matrix of A is

$$\mathbf{G} = \mathbf{A}^T \mathbf{A}$$

- A has m rows and p columns; its Gram matrix G has shape $p \times p$
- Every entry of G is, at its core, the inner product of two columns of A

Figure 1 · The Gram Matrix $A^T A$ of Matrix A

- Columns $a_1, a_2, \dots, a_p$ of A are compressed by ATA into a $p \times p$ inner-product table



The Gram matrix G is symmetric and positive semidefinite

$$G^T = G, \quad \mathbf{x}^T G \mathbf{x} = \|A\mathbf{x}\|^2 \geq 0$$

- For any nonzero p -D vector $\mathbf{x}$, the quadratic form $\mathbf{x}^T G \mathbf{x}$ equals $\|A\mathbf{x}\|^2$ — always nonnegative
- Chapter 13, Section 1 of this book covers positive definiteness in detail

The Gram matrix reveals linear dependence among A's columns

- ① $\det(ATA)=0$ if and only if the columns of A are linearly dependent
- ② $\text{rank}(ATA)=\text{rank}(A)$ — ATA only “compresses” the inner-product relations, without gaining or losing dimensions

Expanding ATA via the first view of matrix multiplication reveals inner products

$$A^T A = [\langle a_1, a_1 \rangle \cdots \langle a_1, a_p \rangle; \vdots \ddots \vdots; \langle a_p, a_1 \rangle \cdots \langle a_p, a_p \rangle]$$

- Diagonal entries: $a_j^T a_j = a_j \cdot a_j = \|a_j\|^2$ — a column's inner product with itself
- Off-diagonal entries: $a_i^T a_j = a_j^T a_i$, which also confirms that ATA is symmetric

Figures 2 & 3 · Diagonal and Off-Diagonal Entries of the Gram Matrix $A^T A$

- Left: diagonal entry $a_1^T a_1 = \|a_1\|^2$; right: off-diagonal entry $a_3^T a_1 = a_1^T a_3$, symmetric across the diagonal

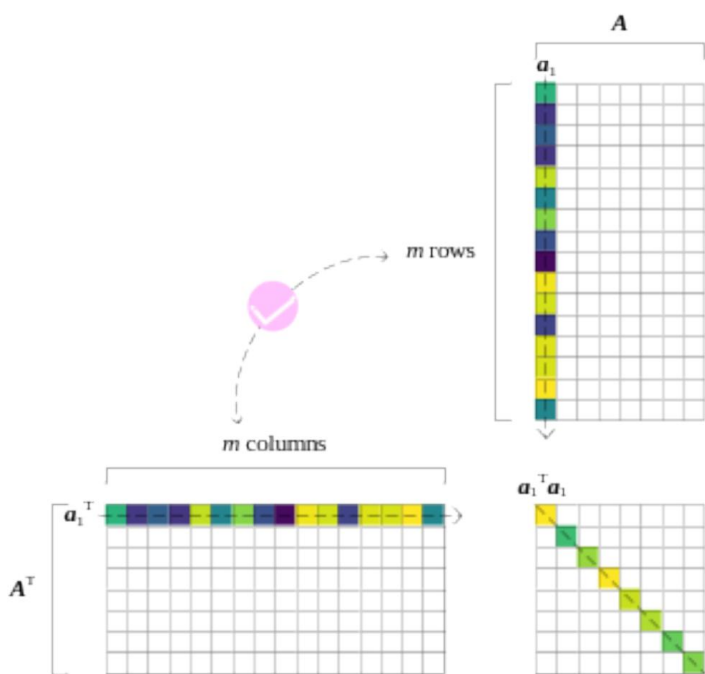


图 2. 矩阵 A 的格拉姆矩阵的对角线元素

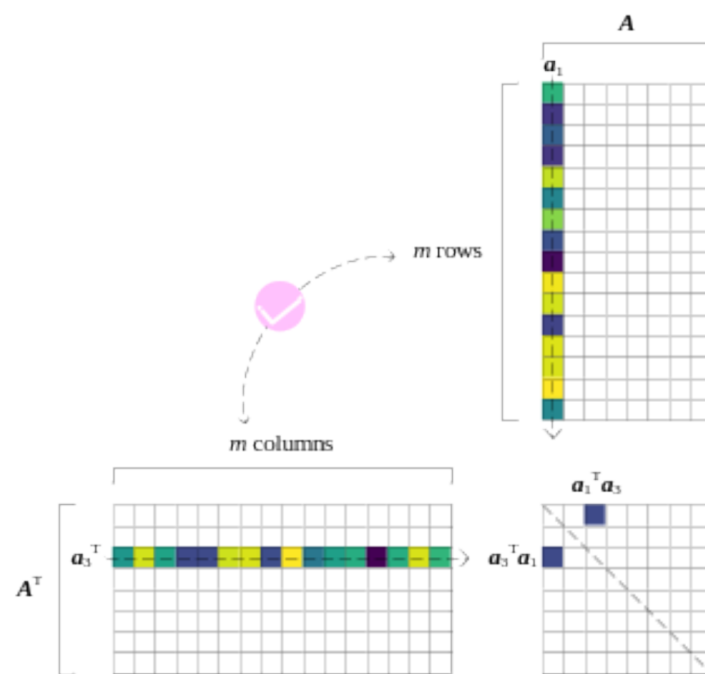


图 3. 矩阵 A 的格拉姆矩阵的非对角线元素

Notably, $A A^T$ is also a Gram matrix

- ① $A A^T$'s main diagonal is each row's inner product with itself — the squared L2 norm of the row
- ② $A A^T$'s off-diagonal entries are pairwise row inner products; $A A^T$ is symmetric too

Figure 4 · The Gram Matrix AA^T of A

- The same matrix A yields AA^T from its columns and A^TA from its rows — both are Gram matrices

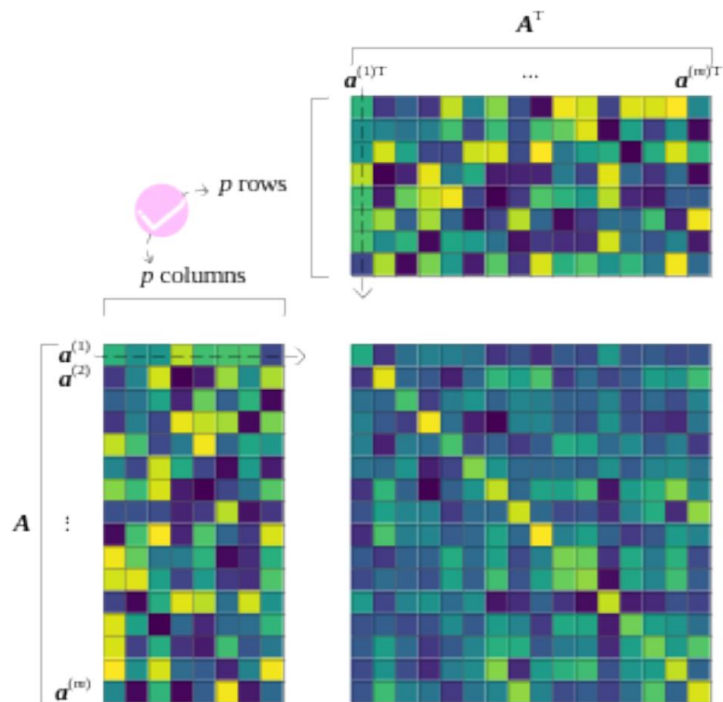


图 4. 矩阵 A^T 的格拉姆矩阵 AA^T

Spectrally decomposing $G=ATA$ gives

$$G = V\Lambda V^T$$

- G is real and symmetric, so V is an orthogonal matrix and Λ is a diagonal matrix

Figure 5 · Spectral Decomposition of the Gram Matrix $A^T A$

- The $p \times p$ matrix G factors into three $p \times p$ matrices: V , Λ , and V^T

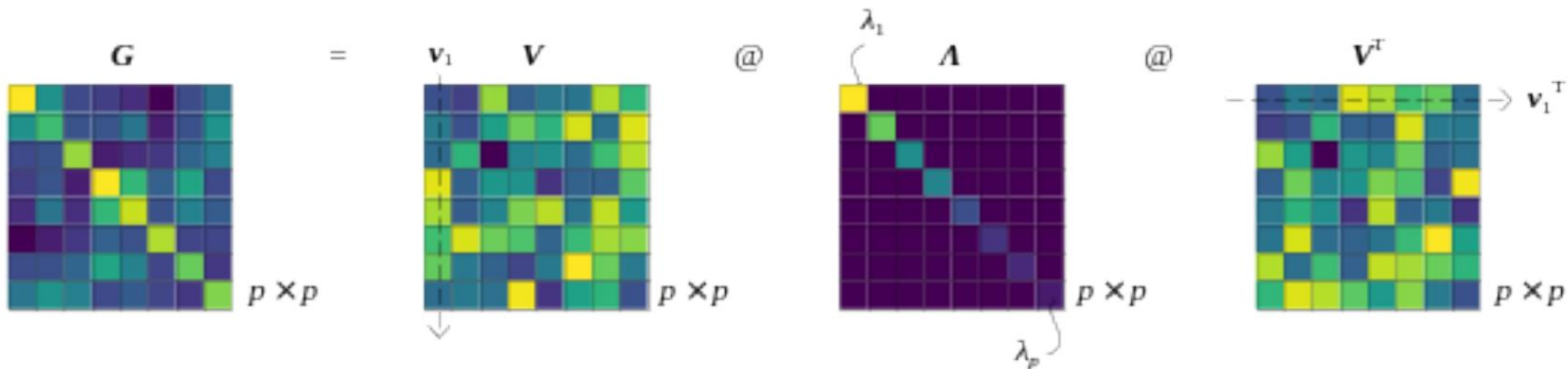


图 5. 对格拉姆矩阵 $A^T A$ 谱分解

Diagonalizing G gives

$$V^T G V = \Lambda$$

- Writing $V=[v_1, v_2, \dots, v_p]$, the (i,j) entry of $V^T G V$ expands to $v_i^T G v_j$

Figure 6 · Diagonalizing the Gram Matrix $A^T A$

- Diagonalization rearranges the product V, G, V^T into a clean diagonal matrix Λ

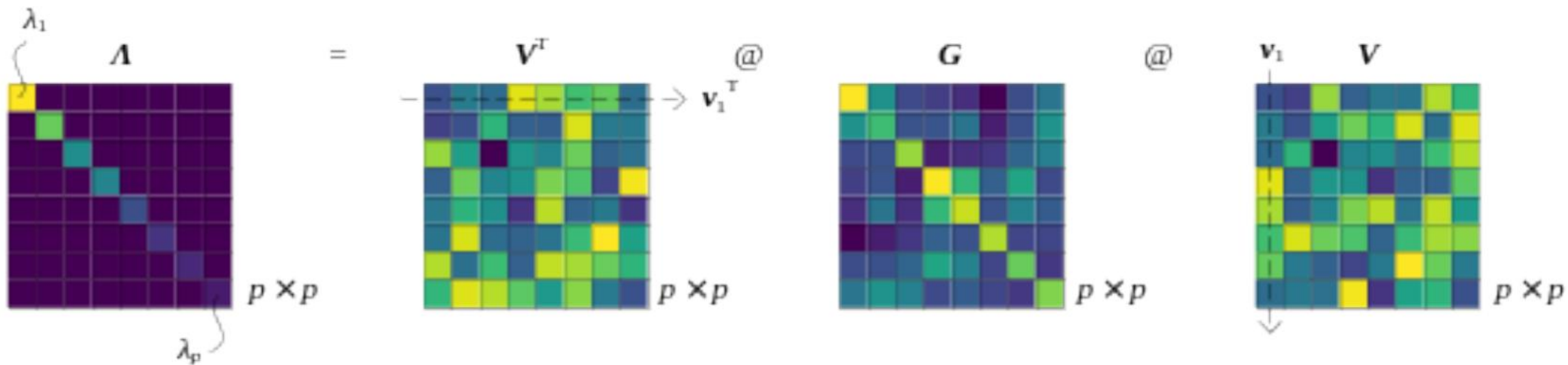


图 6. 格拉姆矩阵 $A^T A$ 对角化

Rearranging gives a pair of identities

$$(Av_i)^T(Av_j) = \lambda_j \quad (i=j), \quad (Av_i)^T(Av_j) = 0 \quad (i \neq j)$$

- $v_i^T G v_j = v_i^T A T A v_j = (Av_i)^T (Av_j)$: the main diagonal ($i=j$) is λ_j , off-diagonal ($i \neq j$) is 0

Figures 7 & 8 · Main-Diagonal and Off-Diagonal Entries of Λ

- Left: $v_1^T G v_1 = \lambda_1$ (main diagonal); right: $v_3^T G v_1 = 0$ (off-diagonal, between two distinct eigenvectors)

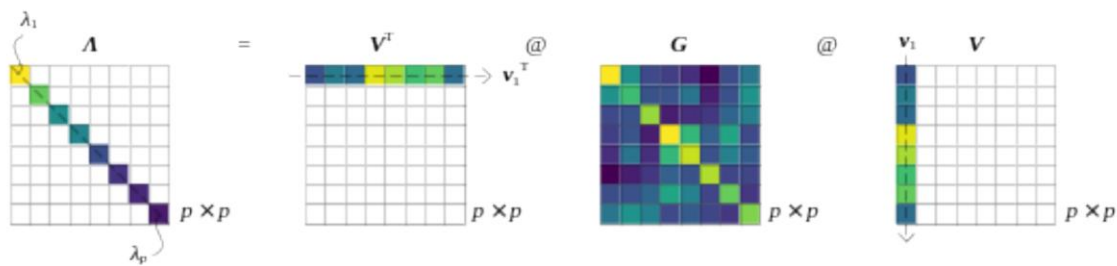


图 7. 对角方阵 Λ 主对角线元素

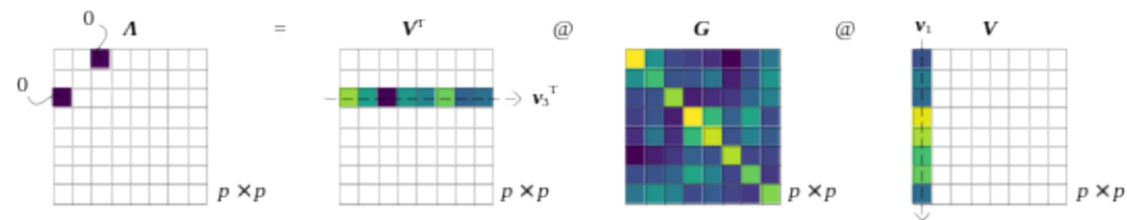


图 8. 对角方阵 Λ 非主对角线元素

Letting $\mathbf{b}_j = \mathbf{A}\mathbf{v}_j$ ($j=1, \dots, p$), the identity above becomes

$$\mathbf{B}^T \mathbf{B} = \mathbf{A}$$

- Since $\mathbf{v}_j$ is a unit vector, $\mathbf{A}\mathbf{v}_j$ is exactly $\mathbf{A}$ projected onto $\text{span}(\mathbf{v}_j)$, giving $\mathbf{b}_j$
- Each row of $\mathbf{A}$ can be viewed as one data point

Figures 9 & 10 · Projecting $\mathbf{A}$ onto $\text{span}(\mathbf{v}_j)$ Gives $\mathbf{b}_j$

- Left: the general case $\mathbf{A}@\mathbf{v}_j=\mathbf{b}_j$; right: a concrete example, $\mathbf{A}$ projected onto $\text{span}(\mathbf{v}_1)$ gives $\mathbf{b}_1$

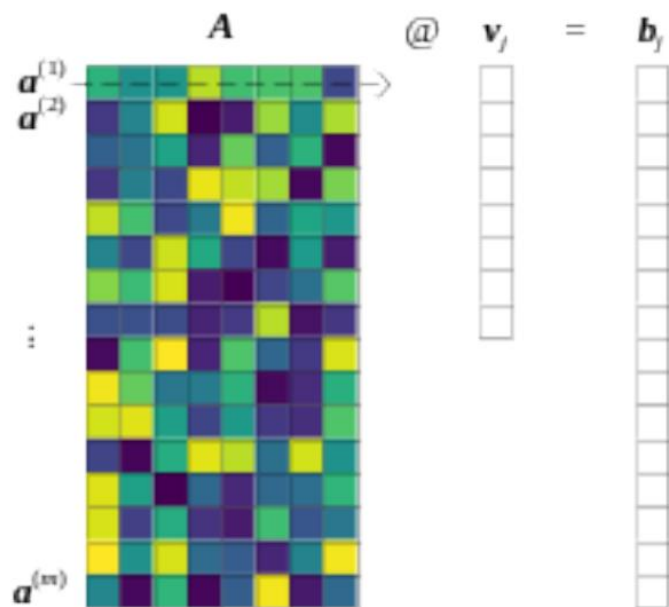


图 9. 矩阵 $\mathbf{A}$ 向 $\text{span}(\mathbf{v}_j)$ 投影结果为 $\mathbf{b}_j$

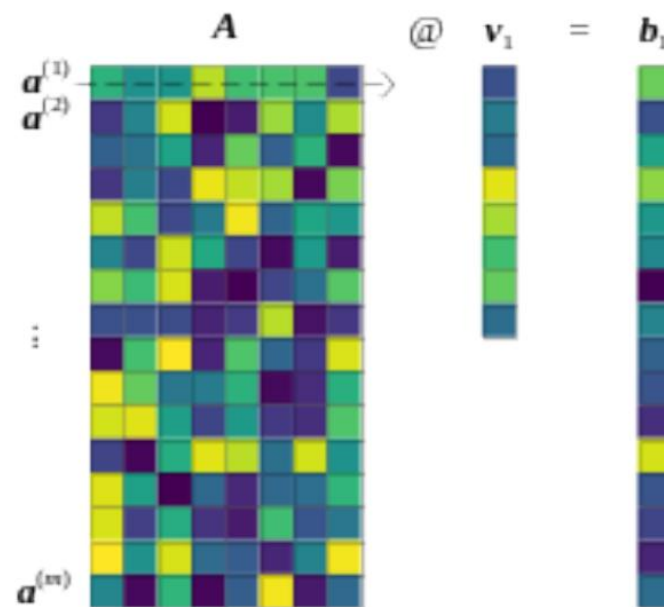


图 10. 矩阵 $\mathbf{A}$ 向 $\text{span}(\mathbf{v}_1)$ 投影结果为 $\mathbf{b}_1$

The product $B=AV$: geometrically, A projected onto $\text{span}(v_1, v_2, \dots, v_p)$ gives B

$$B = AV$$

- The QR decomposition covered earlier in this book is also an orthogonalization — compare the two

Figure 11 · Projecting A onto $\text{span}(v_1, v_2, \dots, v_p)$ Gives B

- Every column of A is projected onto all p eigenvector directions at once, forming B

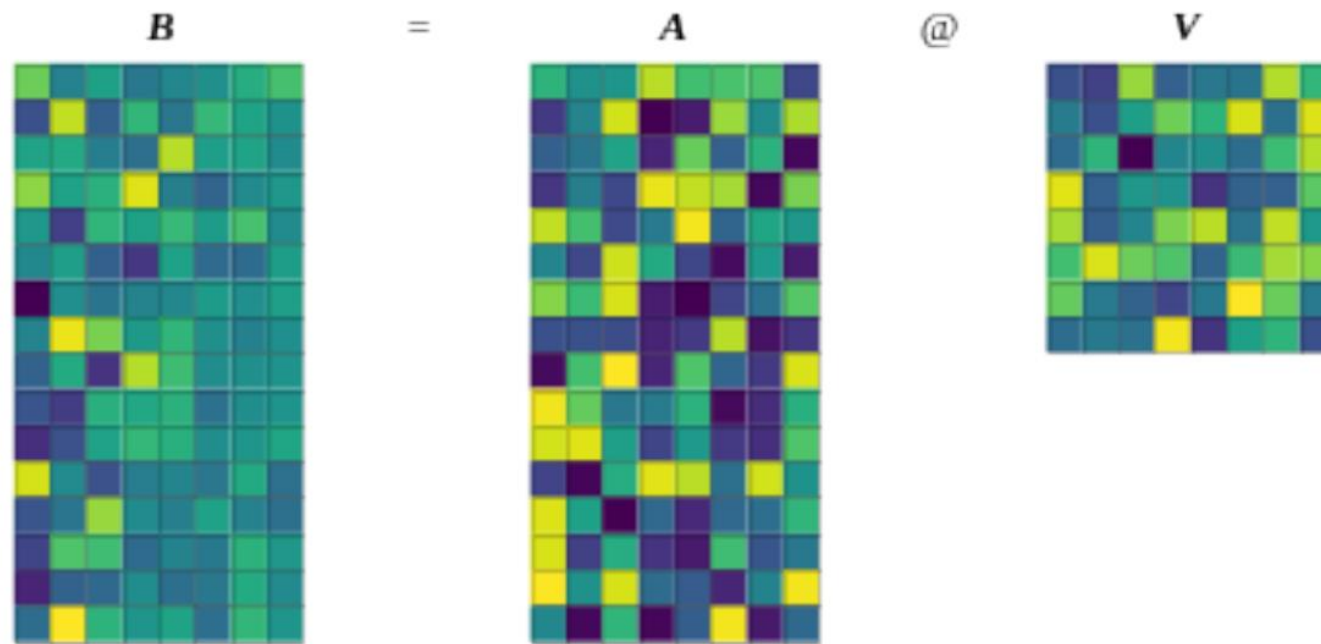


图 11. 矩阵 A 向 $\text{span}(v_1, v_1, \dots, v_p)$ 投影结果为 B

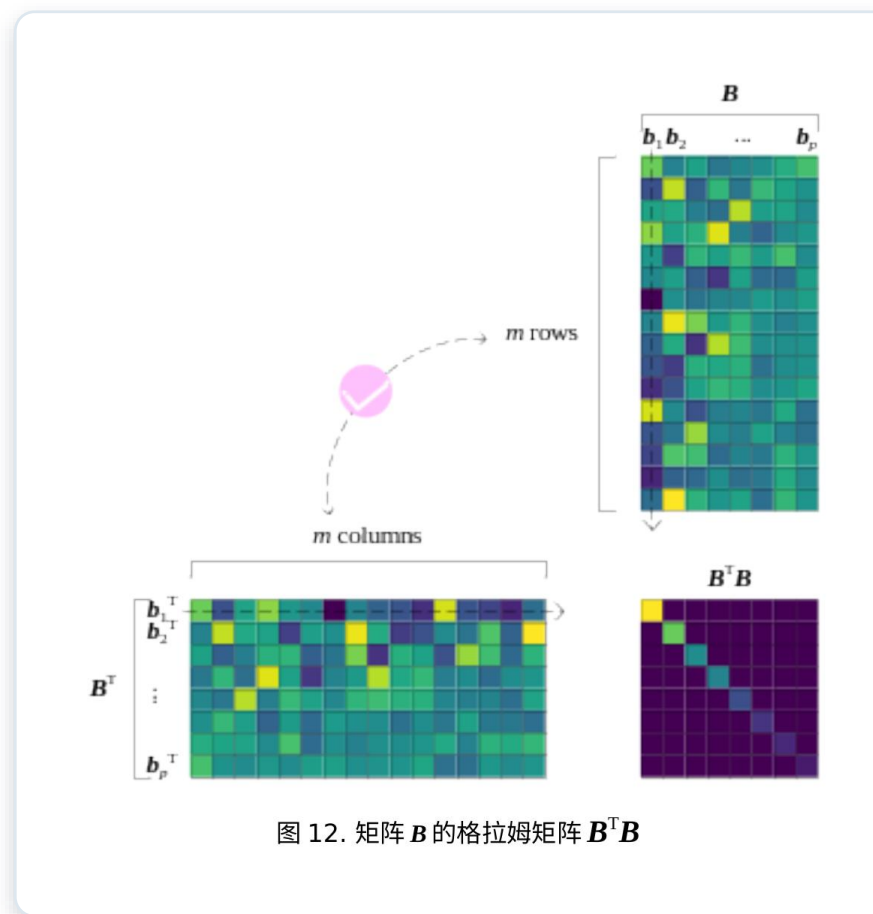
Substituting $B=AV$ into B^TB gives exactly Λ

$$\mathbf{B}^T\mathbf{B} = (\mathbf{AV})^T(\mathbf{AV}) = \mathbf{V}^T\mathbf{A}^T\mathbf{AV} = \mathbf{V}^T\mathbf{GV} = \mathbf{\Lambda}$$

- Notice: B's columns are pairwise orthogonal! $B=AV$ has orthogonalized matrix A

Figure 12 · The Gram Matrix $B^T B$ of B

- $B^T B = \Lambda$ is a diagonal matrix, confirming that B 's columns are pairwise orthogonal



Figures 13 & 14 · Diagonal and Off-Diagonal Entries of B's Gram Matrix

- Diagonal entries $\|b_j\|^2 = \lambda_j$; off-diagonal entries are always 0 — the algebraic meaning of “orthogonal”

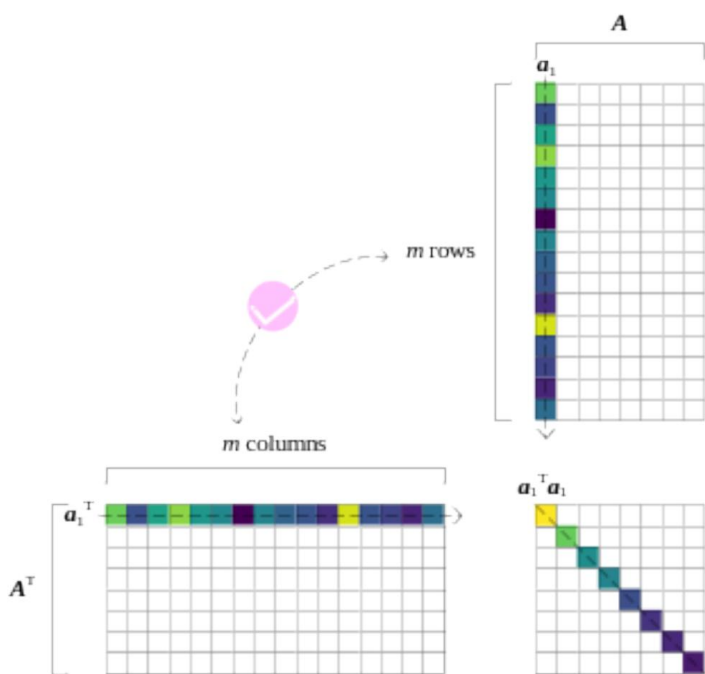


图 13. 矩阵 B 的格拉姆矩阵的对角线元素

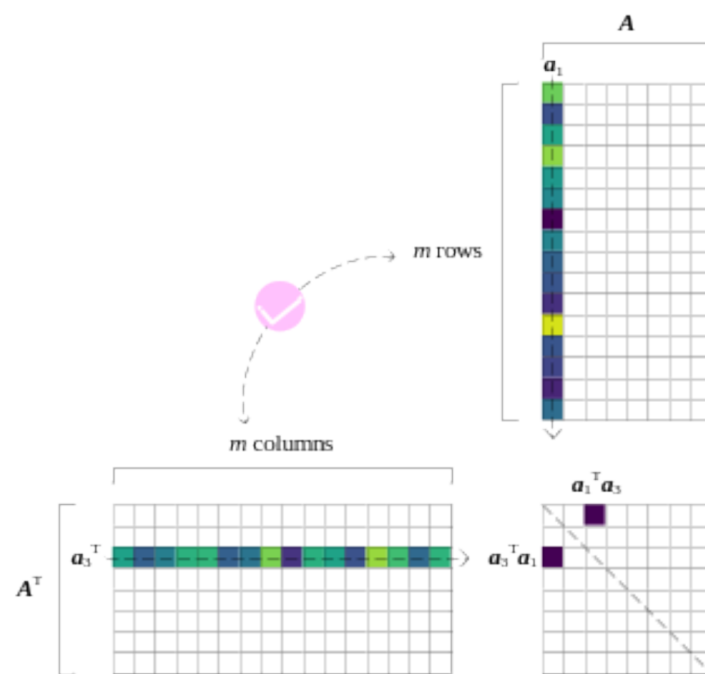
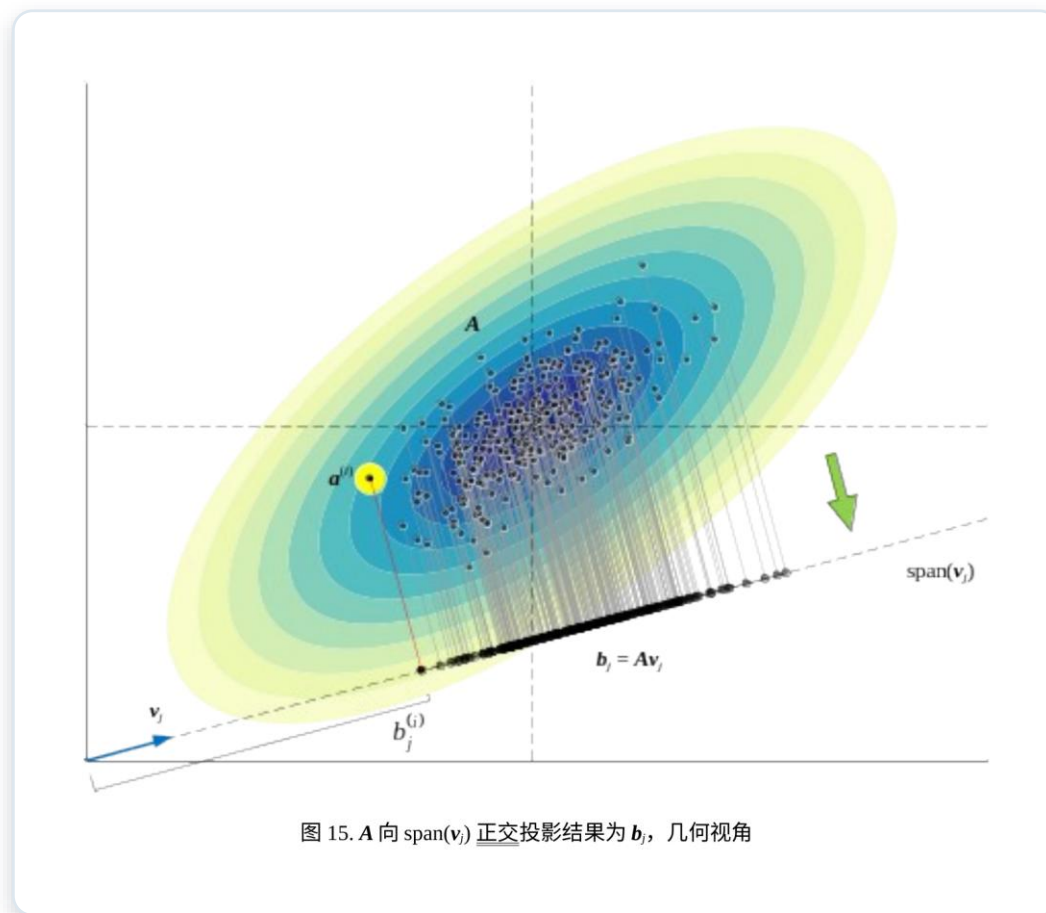


图 14. 矩阵 B 的格拉姆矩阵的非对角线元素

Figure 15 · Projecting A onto $\text{span}(\mathbf{v}_j)$ Gives $\mathbf{b}_j$, Geometrically

- Each row $\mathbf{a}^{(i)}$ of A is a scatter point; its projection onto $\text{span}(\mathbf{v}_j)$ lands at distance $\mathbf{b}_j^{(i)}$



$\|b_j\|^2$ is the sum of squared (signed) distances $b_j(i)$ from the origin

$$\|b_j\|^2 = (b_j^{(1)})^2 + (b_j^{(2)})^2 + \dots = \lambda_j$$

- If $\lambda_1 \geq \lambda_2 \geq \dots$, the eigenvectors v_j also have a clear order of importance
- The ellipse in Figure 15 is really a Mahalanobis distance — detailed in Chapter 13, Section 4

Figure 16 · Eigenvalues and s_j (Their Square Roots) Set an Eigenvector's “Explanatory Power”

- $\|b_1\|^2$ is largest when A projects onto v_1 — the core idea behind Chapter 12's PCA

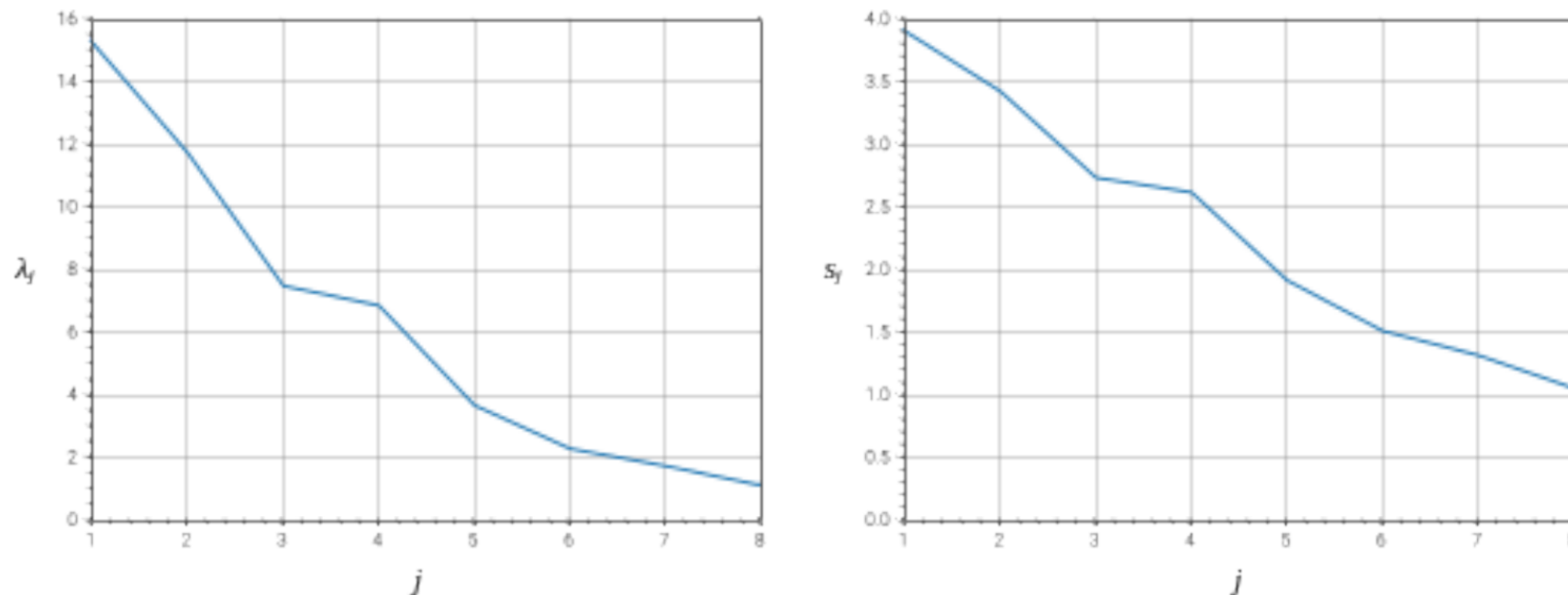


图 16. 特征值、 s_j (特征值平方根) 决定了特征向量的“解释力”大小

UNITIZING

Taking the square root and unitizing $\mathbf{b}_j$ prepares for “stacking the layers”

① Square root: $\|\mathbf{b}_j\| = \sqrt{\lambda_j}$, call it s_j , so $\|\mathbf{b}_j\| = s_j$

② Unitizing: if $\lambda_j > 0$, let $\mathbf{u}_j = \mathbf{b}_j / s_j$, so $\mathbf{b}_j = s_j \mathbf{u}_j$

③ Substituting into $\mathbf{b}_j = \mathbf{A} \mathbf{v}_j$ gives $\mathbf{A} \mathbf{v}_j = s_j \mathbf{u}_j$ — a bridge linking $\mathbf{A}$, $\mathbf{v}_j$, and $\mathbf{u}_j$

Since V is orthogonal, matrix A can be written as a sum of rank-one matrices

$$A = Av_1v_1^T + Av_2v_2^T + \dots = s_1u_1v_1^T + s_2u_2v_2^T + \dots$$

- Every $v_jv_j^T$ is a projection matrix; s_j (the square root of λ_j) sets its “explanatory power”

Figure 17 · Stacking the Layers

- Summing p rank-one matrices $Av_jv_j^T$, each the same shape, exactly rebuilds matrix A

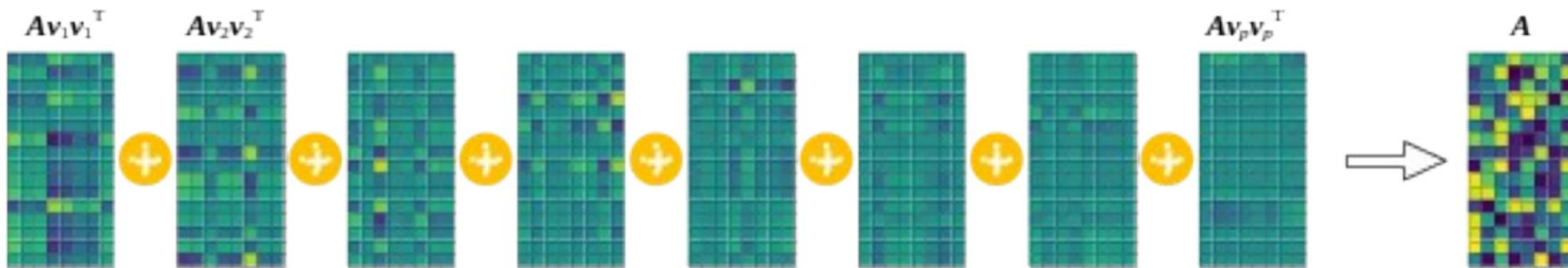


图 17. 层层叠加

Writing the series of products $Av_j = s_j u_j$ ($j=1, \dots, p$) in matrix form

$$AV = US$$

- U's columns are $u_1, \dots, u_p$; S is diagonal, with main-diagonal entries $s_1, \dots, s_p$

$AV=US$

Figure 18 · $AV=US$

- A right-multiplied by V equals U right-multiplied by diagonal S — one matrix identity

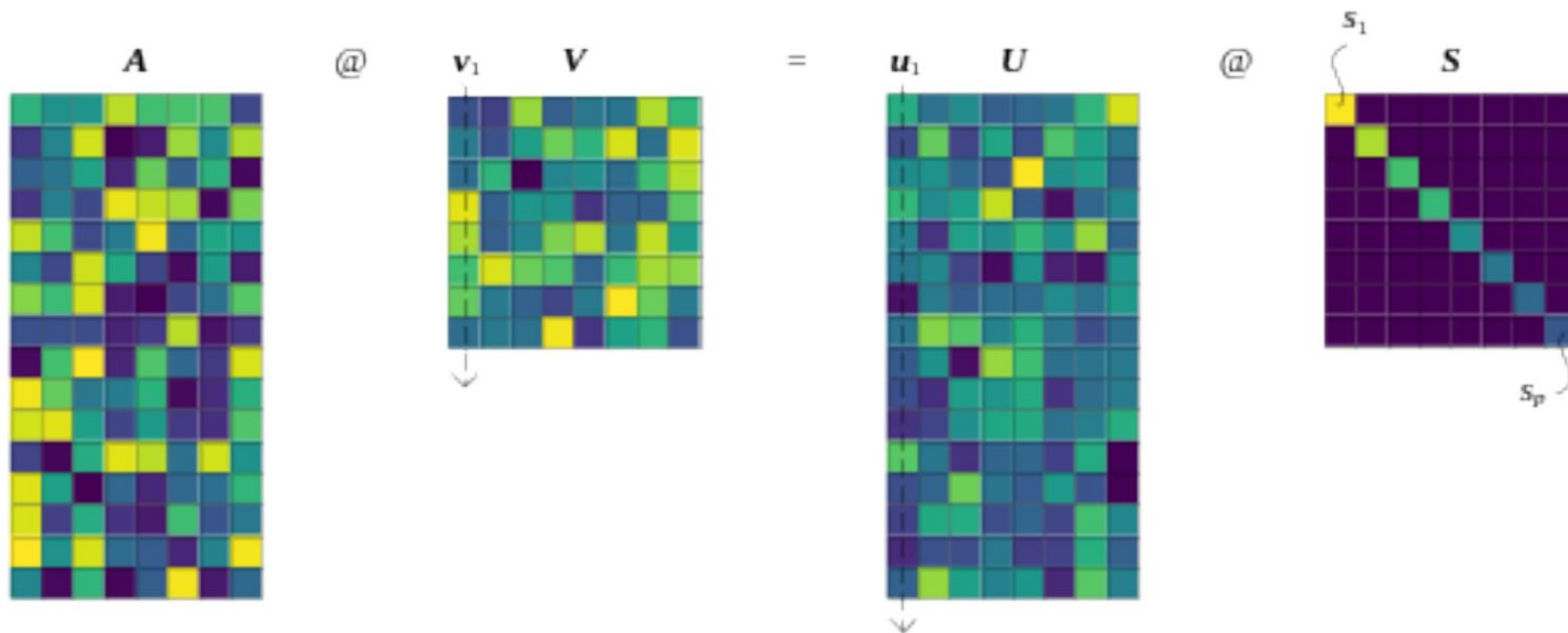


图 18. $AV = US$

SINCE V IS ORTHOGONAL, $V^{-1}=V^T$

Rearranging $AV=US$ gives

$$A = USV^T$$

- This is exactly the singular value decomposition (SVD) from Chapter 14; S 's diagonal entries are the singular values

Figure 19 · The SVD of Matrix A

- You will soon notice: U's columns are also unit vectors, and pairwise orthogonal

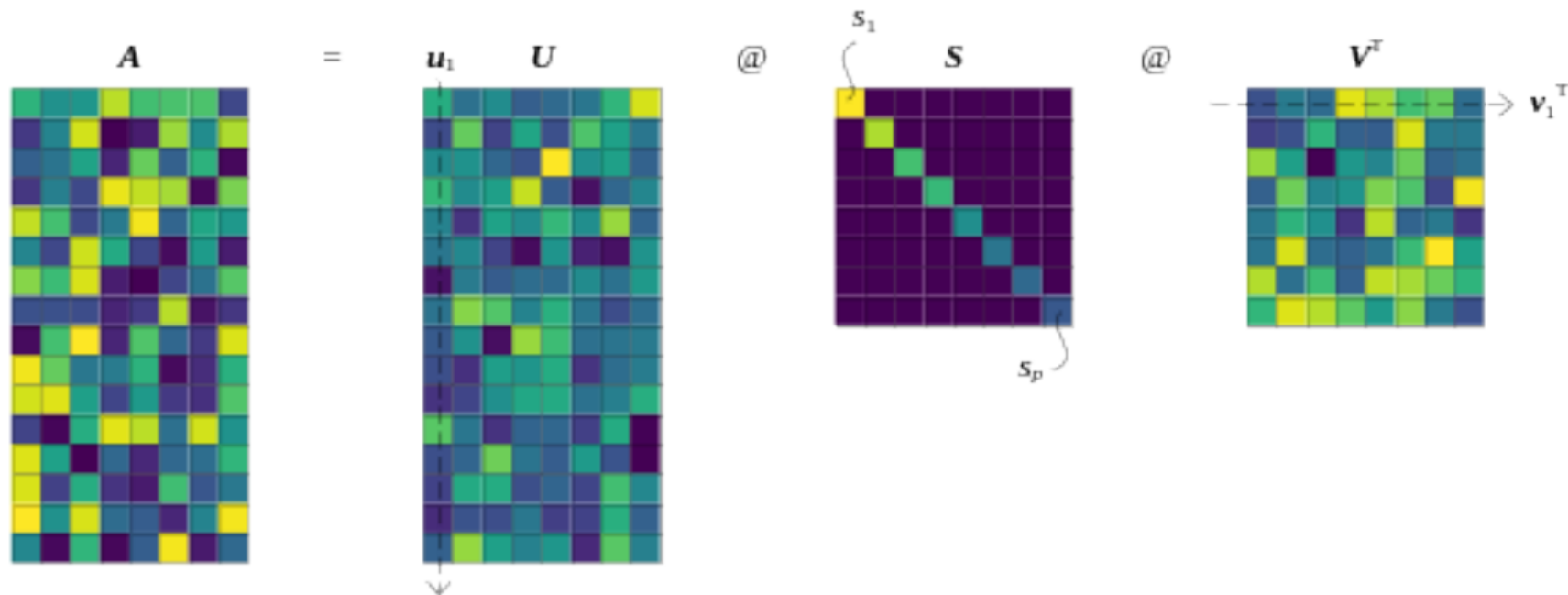


图 19. 矩阵 A 的 SVD 分解

The Gram Matrix Threads Through Many of This Book's Main Ideas

- ① The Gram matrix $G=ATA$ is always symmetric and PSD; its entries are column inner products
- ② Spectral decomposition $G=V\Lambda V^T$: after diagonalizing, each λ_j is a projection direction's “explanatory power”
- ③ $B=AV$ orthogonalizes A — another path to orthogonalization besides QR decomposition
- ④ Stacking layers $A=\sum s_j u_j v_j^T$ leads directly to $A=USVT$ — the singular value decomposition
- ⑤ The most common Gram matrix in practice — the covariance matrix — is the star of the next chapter

Key Terms in This Section

Gram Matrix $G=ATA$; entries are pairwise inner products of A 's columns

Squared L2 Norm The Gram matrix's diagonal entries — a column's inner product with itself

Projection $\mathbf{b}_j = \mathbf{A}\mathbf{v}_j$ $\mathbf{A}$ projected onto $\text{span}(\mathbf{v}_j)$; its squared length equals λ_j

Explanatory Power A larger λ_j (or s_j) means its eigenvector direction carries more information

Singular Value Decomposition (SVD) $A=USVT$; S 's diagonal holds the singular values, U and V are orthogonal

Positive Semidefinite For any nonzero x , $xTGx=\|Ax\|^2 \geq 0$

Spectral Decomposition $G=V\Lambda V^T$, the eigendecomposition of real symmetric G

Orthogonalization $\mathbf{B}=\mathbf{A}\mathbf{V}$ Projecting $\mathbf{A}$ onto $\mathbf{V}$'s column space gives $\mathbf{B}$ with pairwise orthogonal columns

Stacking the Layers $A=\sum s_j \mathbf{u}_j \mathbf{v}_j^T$ decomposes A into a sum of p rank-one matrices

Covariance Matrix One of the most common Gram matrices in practice — detailed next chapter

FURTHER READING

Further Reading

Review · 11.2 Spectral Decomposition

The Gram matrix's spectral decomposition directly applies real-symmetric spectral decomposition

Ahead · 12.1 Data Matrix

The next chapter introduces the covariance matrix — the most common Gram matrix

Preview · 14.1 Singular Value Decomposition

This section's layer-stacking derivation is the most intuitive path into the SVD

Open-Source Repository

All code and figures: <https://github.com/Visualize-ML>

Exercises Q1 – Q9

Q1a Compute the Gram matrix ATA of the matrix $\begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$.

Q2 Spectrally decompose each Gram matrix from Q1: find V and Λ , and verify $G=V\Lambda V^T$.

Q4 Prove that for any nonzero vector x , $x^TATAx=\|Ax\|^2\geq 0$, i.e., ATA is positive semidefinite.

Q6 Explain why $\det(ATA)=0$ iff A 's columns are linearly dependent; verify with a concrete example.

Q8 Compute AAT for both matrices in Q1; compare shape and trace with ATA .

Q1b Compute the Gram matrix ATA of the matrix $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$.

Q3 Prove that for any matrix A , the Gram matrix ATA is always symmetric.

Q5 Write a Python function `gram(A)` that returns ATA , and test it on both matrices from Q1.

Q7 Prove that $\text{rank}(ATA)=\text{rank}(A)$, and check whether the ranks match using the matrices in Q1.

Exercises Q10 – Q18

Q10 Use `numpy.linalg.eigh` to spectrally decompose the Gram matrices from Q1, and compare with your hand calculations.

Q12 Use Python to compute $B=AV$, then print BTB and confirm its off-diagonal entries are near 0.

Q14 For the matrices in Q1, compute $s_j=\sqrt{\lambda_j}$ and unitize b_j to get u_j .

Q16 Write A as the layered sum $\sum s_j u_j v_j^T$, and use Python to verify the sum reproduces A .

Q18 Use `matplotlib` to visualize Figure 15's geometric view — scatter points, projection direction, and results.

Q11 State the definition of $B=AV$, and prove B 's columns are pairwise orthogonal (BTB is diagonal).

Q13 Explain the geometric meaning of $\|b_j\|^2=\lambda_j$; verify it with b_j and λ_j from Q1's matrices.

Q15 Verify $Av_j=s_j u_j$ holds for the matrices in Q1, checking each entry with Python.

Q17 Explain why $v_j v_j^T$ is a projection matrix, and verify idempotence $(v_j v_j^T)^2=v_j v_j^T$.

Exercises Q19 – Q25

Q19 Write $AV=US$ in matrix form, and use Python to confirm US indeed equals AV .

Q21 Use `numpy.linalg.svd` on the matrices from Q1; compare S 's diagonal with this section's s_j values.

Q23 Explore: do ATA and AAT share the same nonzero eigenvalues? Verify with the matrices from Q1.

Q25 Preview: beyond non-symmetric EVD, what other decompositions exist? Guess how Cholesky decomposition relates to the Gram matrix.

Q20 Derive $A=USVT$, and explain why this step requires V to be orthogonal.

Q22 Verify U 's columns are unit vectors and pairwise orthogonal (i.e., $UTU=I$).

Q24 Explain why the covariance matrix is also a kind of Gram matrix — which original matrix does it come from?

10 Prompts to Explore with AI

1 Explain what a Gram matrix is, and why it is always symmetric and positive semidefinite.

2 Explain the geometric meaning of a Gram matrix's diagonal and off-diagonal entries.

3 Explain the relationship and difference between ATA and AAT , and how their nonzero eigenvalues relate.

4 Explain what V and Λ represent in the Gram matrix's spectral decomposition $G=V\Lambda V^T$.

5 Explain why $B=AV$ orthogonalizes matrix A , and how this compares to QR decomposition.

10 Prompts to Explore with AI

6 Explain why $\|b_j\|^2$ exactly equals the eigenvalue λ_j , and the geometric meaning behind it.

7 Explain how eigenvalue size determines an eigenvector's “explanatory power,” and its link to PCA.

8 Explain how to write matrix A as the layered sum $s_1 u_1 v_1^T + s_2 u_2 v_2^T + \dots$

9 Derive the full path from $AV=US$ to $A=USVT$, and explain why this is the singular value decomposition.

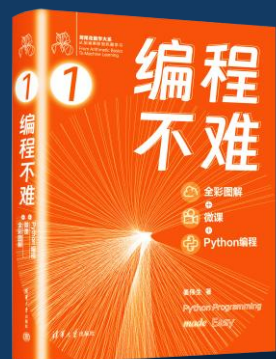
10 Use Python to demonstrate, step by step, how the Gram matrix's spectral decomposition leads to the SVD.

数学

数学基础

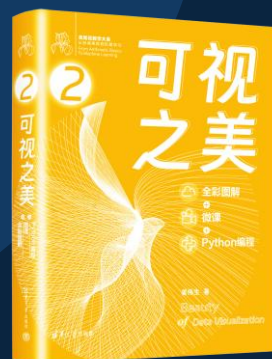
线性代数

概率统计

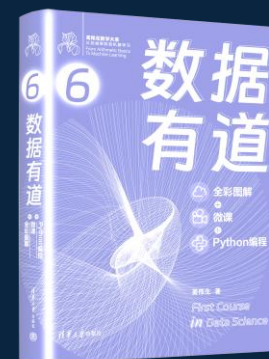
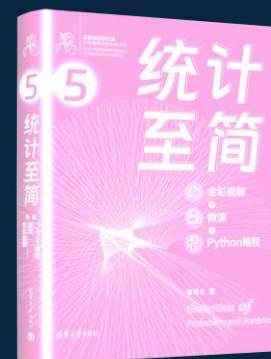
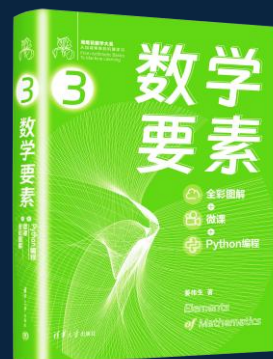


Python编程

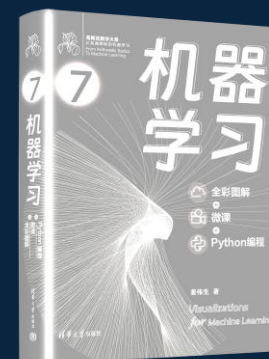
工具



数据可视化



数据分析
图和网络



回归、降维
分类、聚类

实践

